

Appendix A Selection of Sample Solutions with Different Seeds

A.1 Sample Solution where the Number of Candidates = 60

A.1.1 Sample output simulation results, seed=1 candidates = 60

RL + NSGA-III simulation profile is reported in Table A1.

Table A1: RL + NSGA-III simulation profile

Generation	Best Coverage	epsilon
0	0.8683	0.392
10	0.9438	0.320
20	0.9837	0.262
30	0.9911	0.214
40	0.9985	0.175
50	0.9985	0.143
60	0.9985	0.117
70	0.9985	0.095
80	0.9985	0.078
90	0.9985	0.064
100	0.9985	0.052
110	0.9985	0.042
120	0.9985	0.035
130	0.9985	0.028
140	0.9985	0.023

Table A2: MILP Pareto approximation

Coverage	Energy(J)	Cost	Sensors	Connected
0.9911	0.032640	263.89	26	True
0.9749	0.028424	233.23	23	True
0.9719	0.029318	216.93	23	True
0.9601	0.026593	209.56	22	True
0.9320	0.023444	190.30	20	True
0.9305	0.023316	189.43	20	True
0.9038	0.021453	180.94	19	True
0.8580	0.019689	161.40	17	True
0.7663	0.015622	145.50	15	True
0.2899	0.002824	48.49	5	True
0.2426	0.002308	35.17	4	True
0.2027	0.001654	28.00	3	True
0.2012	0.001836	27.04	3	True
0.1923	0.001681	26.04	3	True

Table A3: RL + NSGA-III Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9985	0.041857	332.16	32	True
0.9867	0.033757	283.88	28	True
0.9808	0.031558	273.66	27	True
0.9541	0.027048	205.76	22	True
0.9541	0.027048	205.76	22	True
0.9497	0.026432	239.53	24	True
0.9497	0.026432	239.53	24	True
0.9393	0.025007	197.56	21	True
0.9393	0.025007	197.56	21	True
0.9393	0.025007	197.56	21	True
0.9393	0.025007	197.56	21	True
0.9349	0.024805	231.33	23	True
0.9127	0.023500	190.04	20	True
0.9083	0.022942	222.84	22	True
0.9024	0.021443	185.72	19	True

Referring to Table A5, MILP achieves approximately $2.44\times$ the hypervolume of RL+NSGA-III and $7.74\times$ that of Random Search. RL+NSGA-III achieves approximately $3.17\times$ the hypervolume of Random Search.

Execution Time

MILP weight sweep : 123.139 s, RL + NSGA-III : 97.758 s, Random Search : 139.692 s

Table A4: Random search Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9689	0.045079	342.80	33	True
0.9601	0.047539	324.96	30	True
0.9571	0.039692	329.42	30	True
0.9541	0.039350	323.95	30	True
0.9467	0.039724	289.11	27	True
0.9453	0.036571	330.41	31	True
0.9393	0.031173	267.23	25	True
0.9098	0.031041	266.82	26	True
0.8994	0.030471	295.08	26	True
0.8979	0.030582	250.98	23	True
0.8964	0.033652	239.72	23	True
0.8920	0.029381	269.94	24	True
0.8891	0.031610	250.89	23	True
0.8876	0.028759	254.32	24	True
0.8787	0.029923	251.86	24	True

Table A5: Hypervolume Comparison

Method	Normalized Hypervolume	Relative to Random
MILP	0.2983	7.74×
RL + NSGA-III	0.1222	3.17×
Random Search	0.0385	1.00×

In terms of computational efficiency, RL + NSGA-III achieved the lowest execution time at 97.758 s, compared with 123.139 s for the MILP weight sweep and 139.692 s for Random Search. Consequently, RL + NSGA-III was approximately 1.26 times faster than MILP, corresponding to a 20.61% reduction in execution time, and approximately 1.43 times faster than Random Search, corresponding to a 30.02% reduction. These results demonstrate that, for this instance, the proposed RL + NSGA-III framework provides the most computationally efficient solution approach among the three methods.

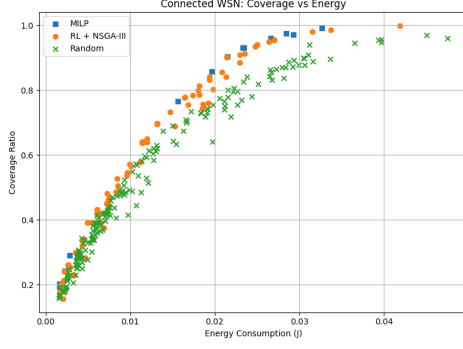
RL operator usage:

Bit Flip: 1326, Add Sensor : 605, Remove Sensor: 6298, Swap Sensor: 383,
Coverage Guided: 6388

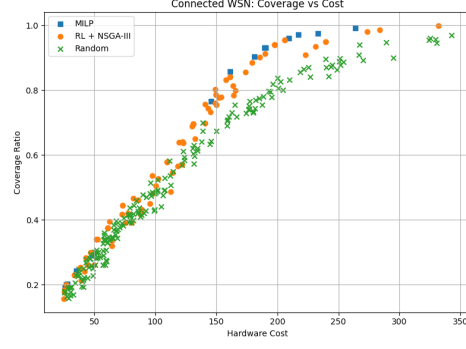
Example RL + NSGA-III Solution

Coverage = 0.9985, Energy = 0.041857 J, Hardware Cost = 332.16, Active Sensors = 32, Connected = True

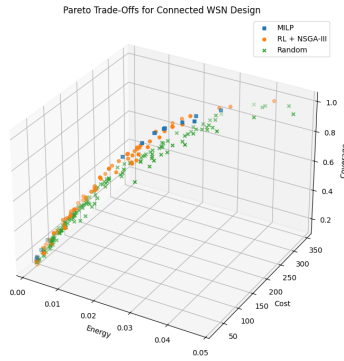
The three tables (Table A2, Table A3, and Table A4) and their corresponding plots (Figure A1a, Figure A1b, Figure A1c, Figure A1d) show a clear hierarchy. All reported solutions satisfy the connectivity constraint, so the comparison can focus on the three optimization objectives: maximize coverage while minimizing energy consumption and hardware cost. Sensor count provides an additional measure of deployment efficiency.



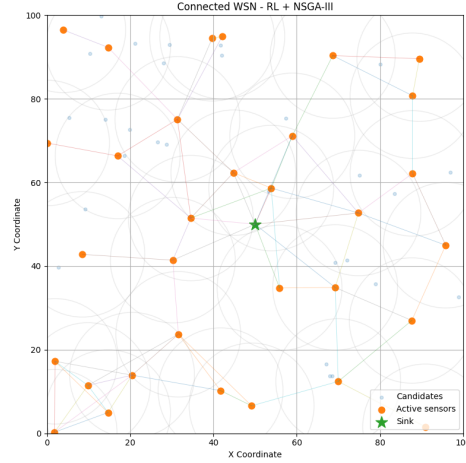
(a) Connected WSN: Coverage vs Energy



(b) Connected WSN: Coverage vs Cost



(c) Pareto Trade-Offs for connected WSN design



(d) Connected WSN coordinates - RL + NSGA-III

Fig. A1: Overall comparison of the four experimental results for seed =1, candidates=60.

Overall Comparison

The three Pareto fronts show a clear performance hierarchy. MILP provides the strongest overall trade-off frontier, RL + NSGA-III provides competitive high-coverage solutions but with weaker diversity, while Random Search is substantially less efficient in energy, cost, and sensor usage.

The results indicate that MILP provides the widest Pareto trade-off spectrum, ranging from extremely inexpensive low-coverage configurations to configurations exceeding 99% coverage. RL+NSGA-III concentrates much more heavily on the high-coverage region, whereas random search occupies a relatively narrow and inefficient

Table A6: Overall Comparison with Seed=1

Criterion	MILP	RL + NSGA-III	Random Search
Maximum coverage	0.9911	0.9985	0.9689
Minimum coverage represented	0.1923	0.9024	0.8787
Energy range (J)	0.001654–0.032640	0.021443–0.041857	0.028759–0.047539
Cost range	26.04–263.89	185.72–332.16	239.72–342.80
Sensor range	3–26	19–32	23–33
Reported solutions	14	15	15
Unique solutions	14	10	15
All connected	Yes	Yes	Yes

region.

Overall Ranking

For these particular experimental results, the methods can be characterized as: Resource efficiency and Pareto diversity

$$\boxed{\text{MILP} > \text{RL+NSGA-III} \gg \text{Random}}$$

Maximum achievable coverage

$$\boxed{\text{RL+NSGA-III} > \text{MILP} > \text{Random}}$$

Overall practical multi-objective performance

$$\boxed{\text{MILP} \approx \text{RL+NSGA-III} \gg \text{Random}}$$

MILP is preferable when cost, energy, and sensor economy are important, whereas RL+NSGA-III becomes particularly attractive when maximizing coverage is prioritized or when the eventual objective is to scale the methodology to instances where exact/Mathematical Programming approaches may become computationally demanding. The random algorithm serves well as a baseline but produces no competitive point relative to the two optimization-based approaches.

Overall Conclusion

Both MILP and RL+NSGA-III significantly outperform random search across all metrics, with every random-search solution strictly dominated by both optimized approaches. While RL+NSGA-III achieves the absolute highest coverage (99.85%), it requires 32 sensors and higher resource usage. In contrast, MILP delivers a nearly identical peak coverage of 99.11% using only 26 sensors—saving roughly 28% in energy and 26% in cost—while providing the broadest, most resource-efficient Pareto front overall. At a fixed 96.01% coverage level, both methods reduce energy by $\approx$

44%, cost by 36%, and sensor count by $\approx 27\%$ compared to random search. Ultimately, MILP offers the best overall efficiency, while RL+NSGA-III serves as a strong alternative tailored for maximum coverage.

A.1.2 Sample output simulation results, seed=3 candidates = 60

RL + NSGA-III simulation profile is reported in Table A7.

Table A7: RL + NSGA-III simulation profile

Generation	Best Coverage	epsilon
0	0.7559	0.392
10	0.8595	0.320
20	0.9098	0.262
30	0.9127	0.214
40	0.9127	0.175
50	0.9127	0.143
60	0.9127	0.117
70	0.9127	0.095
80	0.9009	0.078
90	0.8802	0.064
100	0.8772	0.052
110	0.8846	0.042
120	0.8802	0.035
130	0.8802	0.028
140	0.8772	0.023

Table A8: MILP Pareto approximation

Coverage	Energy(J)	Cost	Sensors	Connected
0.8994	0.030280	242.40	22	True
0.8979	0.028837	255.52	23	True
0.8698	0.025486	223.89	20	True
0.8328	0.021293	201.75	18	True
0.8195	0.021018	185.37	17	True
0.7885	0.019264	171.61	16	True
0.7692	0.017782	166.05	15	True
0.7322	0.015465	157.47	14	True
0.6450	0.011932	128.07	12	True
0.6095	0.010461	119.62	11	True
0.6065	0.011036	113.88	11	True
0.2485	0.002231	42.06	4	True
0.1908	0.001624	27.30	3	True
0.1893	0.001524	28.09	3	True

Table A9: RL + NSGA-III Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.8772	0.028677	252.27	22	True
0.8432	0.022855	238.69	20	True
0.8432	0.022855	238.69	20	True
0.8432	0.022855	238.69	20	True
0.8373	0.025796	212.31	18	True
0.8299	0.028696	208.35	18	True
0.8299	0.028696	208.35	18	True
0.8299	0.028696	208.35	18	True
0.8240	0.021487	215.49	18	True
0.8240	0.021487	215.49	18	True
0.7988	0.031882	197.77	17	True
0.7959	0.024807	194.21	17	True
0.7959	0.024807	194.21	17	True
0.7929	0.021471	197.27	18	True
0.7899	0.026903	193.66	17	True

Referring to Table A11, MILP achieves approximately $3.15\times$ the hypervolume of RL+NSGA-III and $4.26\times$ that of Random Search. RL+NSGA-III achieves approximately $1.35\times$ the hypervolume of Random Search.

Execution Time

MILP weight sweep : 206.981 s, RL + NSGA-III : 93.893 s, Random Search : 144.725 s

Table A10: Random search Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.8713	0.039240	371.33	32	True
0.8624	0.039588	363.15	32	True
0.8609	0.035798	358.44	30	True
0.8550	0.033608	319.72	27	True
0.8462	0.034007	297.16	26	True
0.8462	0.031056	322.37	27	True
0.8432	0.034884	271.27	24	True
0.8417	0.031021	304.41	25	True
0.8373	0.027202	284.13	24	True
0.8299	0.032794	256.07	24	True
0.8269	0.026130	258.91	23	True
0.8210	0.031894	250.89	22	True
0.8151	0.025832	260.62	22	True
0.8047	0.031475	242.48	21	True
0.8003	0.021655	225.50	18	True

Table A11: Hypervolume Comparison

Method	Normalized Hypervolume	Relative to Random
MILP	0.3623	4.26×
RL + NSGA-III	0.1151	1.35×
Random Search	0.0851	1.00×

These results indicate that RL + NSGA-III provides the highest computational efficiency, completing the optimization considerably faster than both MILP and Random Search. In particular, its more than 54% runtime reduction relative to MILP is significant because RL + NSGA-III is designed to generate a multi-objective Pareto approximation rather than merely sample solutions randomly. Therefore, for this experiment, the hybrid RL + NSGA-III framework offers a particularly favorable compromise between optimization capability and computational cost, whereas MILP incurs the highest computational burden.

RL operator usage:

Bit Flip : 666, Add Sensor : 685

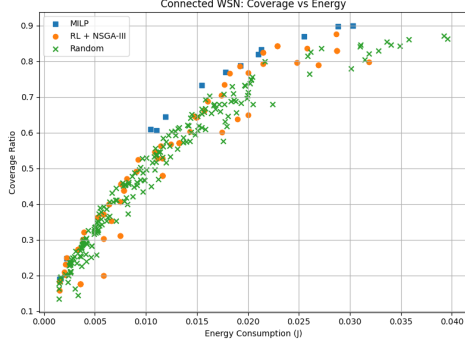
Remove Sensor : 2902, Swap Sensor : 835, Coverage Guided : 9912

Example RL+ NSGA-III Solution

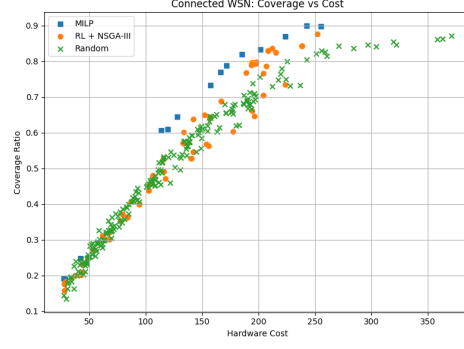
Coverage = 0.8772, Energy = 0.028677 J, Hardware Cost = 252.27, Active Sensors = 22, Connected = True

Overall Comparison

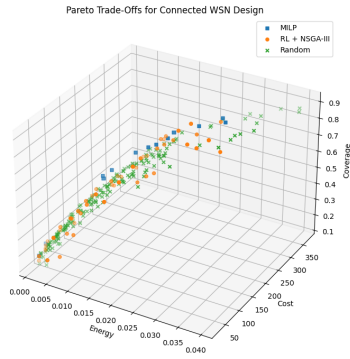
The three Pareto fronts show a clear performance hierarchy. MILP provides the strongest overall trade-off frontier, RL + NSGA-III provides competitive high-coverage solutions but with weaker diversity, while Random Search is substantially less efficient in energy, cost, and sensor usage. For ample comparison details, refer to the comparison Figure A2, Table A8, Table A9, and Table A10.



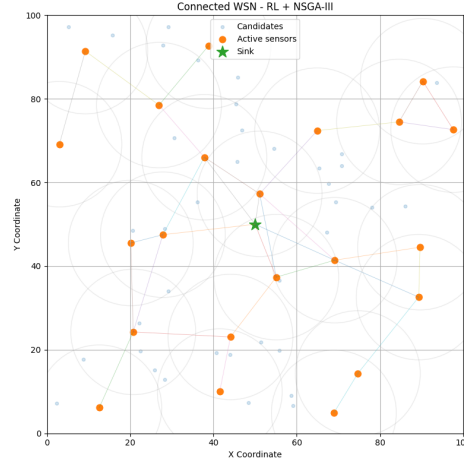
(a) Connected WSN: Coverage vs Energy



(b) Connected WSN: Coverage vs Cost



(c) Pareto Trade-Offs for connected WSN design



(d) Connected WSN coordinates - RL + NSGA-III

Fig. A2: Overall comparison of the four experimental results for seed=3, candidate=60.

A particularly important observation is that all reported solutions are connected, meaning that all three approaches successfully satisfy the connectivity constraint. Consequently, the main differences concern coverage, energy consumption, cost, and number of deployed sensors.

Overall interpretation

The results suggest the ranking

$$\boxed{\text{MLP} > \text{RL} + \text{NSGA-III} \gg \text{Random Search}}$$

Table A12: Overall Comparison with Seed=3

Criterion	MILP	RL + NSGA-III	Random Search
Maximum coverage	0.8994	0.8772	0.8713
Minimum energy (J)	0.001524	0.021471	0.021655
Minimum cost	27.30	193.66	225.50
Minimum sensors	3	17	18
Maximum sensors	23	22	32
Unique solutions	14/14	9/15	15/15
Connectivity	100%	100%	100%

for this particular experiment. MILP is the strongest method, because it reaches the highest coverage, produces the broadest and most diverse trade-off front, and frequently obtains better energy-cost combinations with fewer sensors. RL + NSGA-III is nevertheless highly competitive, particularly in the high-coverage region, and strongly outperforms Random Search. Its main weakness here is its limited Pareto diversity and the repeated convergence to identical solutions. Random Search provides the weakest solutions, often requiring many additional sensors and substantially greater cost and energy to obtain similar or lower coverage.

Overall Conclusion

The most defensible conclusion would be that these results demonstrate the effectiveness of RL + NSGA-III as an approximate multi-objective optimizer relative to Random Search, while MILP remains the superior reference method for this instance size. The principal advantage of RL should therefore be investigated in terms of scalability and execution time on larger WSN instances, where exact or MILP-based optimization is expected to become increasingly computationally expensive.

A.1.3 Sample output simulation results, seed=7 candidates = 60

RL + NSGA-III simulation profile is reported in Table A13.

Table A13: RL + NSGA-III simulation profile

Generation	Best Coverage	epsilon
0	0.7899	0.392
10	0.8905	0.320
20	0.9157	0.262
30	0.9216	0.214
40	0.9083	0.175
50	0.9216	0.143
60	0.9157	0.117
70	0.9024	0.095
80	0.9024	0.078
90	0.8964	0.064
100	0.9157	0.052
110	0.9379	0.042
120	0.9408	0.035
130	0.9438	0.028
140	0.9423	0.023

Table A14: MILP Pareto approximation

Coverage	Energy(J)	Cost	Sensors	Connected
0.9379	0.031699	270.52	24	True
0.9305	0.030089	251.69	23	True
0.9246	0.027794	251.74	22	True
0.9231	0.027802	247.73	22	True
0.8876	0.024469	219.71	20	True
0.8831	0.025319	205.34	19	True
0.8314	0.021592	180.23	17	True
0.8151	0.019528	189.62	17	True
0.7367	0.015734	160.50	14	True
0.7249	0.018337	138.19	14	True
0.6361	0.011622	144.68	12	True
0.2515	0.003291	35.88	4	True
0.1893	0.002103	25.64	3	True
0.1864	0.001513	32.60	3	True

Table A15: RL + NSGA-III Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9364	0.043612	326.61	29	True
0.9216	0.037204	270.38	24	True
0.9127	0.035595	257.67	23	True
0.9112	0.031120	279.45	24	True
0.9098	0.031077	269.53	23	True
0.9068	0.035609	256.38	23	True
0.9053	0.031032	271.20	23	True
0.9053	0.031032	271.20	23	True
0.9038	0.041482	247.75	22	True
0.9038	0.041482	247.75	22	True
0.9009	0.033952	247.01	22	True
0.8743	0.027350	248.62	21	True
0.8683	0.048831	221.70	20	True
0.8654	0.024988	244.61	21	True
0.8639	0.035460	215.63	19	True

Table A16: Random search Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.8950	0.044769	401.96	36	True
0.8920	0.042650	403.18	35	True
0.8905	0.034611	298.72	26	True
0.8876	0.031187	304.32	27	True
0.8817	0.044572	297.96	27	True
0.8802	0.035707	286.66	26	True
0.8728	0.038403	272.04	24	True
0.8713	0.035502	296.33	26	True
0.8698	0.030749	282.14	26	True
0.8654	0.038146	257.86	24	True
0.8609	0.032981	272.10	24	True
0.8521	0.032592	258.75	25	True
0.8491	0.032276	280.19	25	True
0.8491	0.029657	304.36	26	True
0.8462	0.032000	271.28	24	True

Table A17: Hypervolume Comparison

Method	Normalized Hypervolume	Relative to Random
MILP	0.3674	7.40×
RL + NSGA-III	0.1081	2.18×
Random Search	0.0497	1.00×

Referring to Table A17, MILP achieves approximately 3.40× the hypervolume of RL+NSGA-III and 7.40× that of Random Search. RL+NSGA-III achieves approximately 2.18× the hypervolume of Random Search.

Execution Time:

MILP weight sweep :401.513 s, RL + NSGA-III :120.789 s, Random Search :136.577 s

These results provide particularly strong evidence for the computational scalability of RL + NSGA-III. With an execution time of only 120.789 s, it is the fastest of

the three approaches, reducing computation time by nearly 70% relative to MILP and by approximately 11.6% relative to Random Search. Consequently, while MILP may provide highly resource-efficient Pareto solutions, its superior optimization quality comes at a considerable computational cost. RL + NSGA-III offers a much more favorable computational trade-off, combining multi-objective optimization capability with substantially faster execution.

RL operator usage:

Bit Flip : 1548, Add Sensor : 397

Remove Sensor : 2641, Swap Sensor : 381, Coverage Guided : 10033

Example RL + NSGA-III Solution

Coverage = 0.9364, Energy = 0.043612 J, Hardware Cost = 326.61, Active Sensors = 29, Connected = True

The three Pareto fronts show a clear performance hierarchy: MILP provides the strongest overall trade-offs, RL + NSGA-III is substantially better than Random Search and approaches MILP in coverage, while Random Search is consistently the least resource-efficient. All reported solutions satisfy connectivity, so the comparison is driven entirely by coverage, energy, cost, and sensor count. For ample comparison details, refer to the comparison Figure A3, Table A14, Table A15, and Table A16.

Overall Comparison

The average coverage should not be used directly to rank the algorithms because

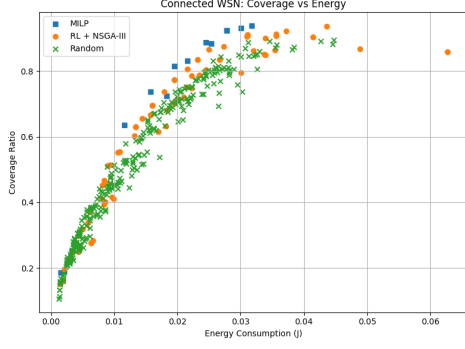
Table A18: Overall Comparison with Seed=7

Criterion	MILP	RL + NSGA-III	Random Search
Maximum coverage	0.9379	0.9364	0.8950
Minimum coverage represented	0.1864	0.8639	0.8462
Minimum energy (J)	0.001513	0.024988	0.029657
Maximum energy (J)	0.031699	0.048831	0.044769
Minimum cost	25.64	215.63	257.86
Maximum cost	270.52	326.61	403.18
Sensor range	3–24	19–29	24–36
Unique listed solutions	14	13	15
Connectivity	100%	100%	100%

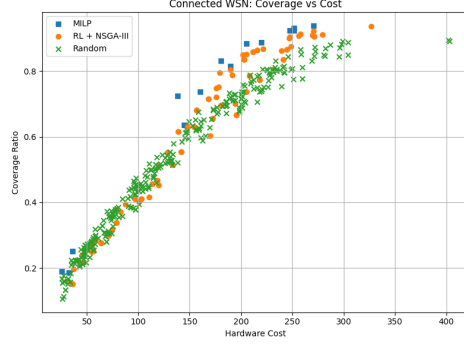
the MILP front deliberately spans a much wider operating range, including very low-cost, low-energy solutions with coverage below 0.30. RL and Random Search are concentrated almost entirely in the high-coverage region.

The Overall Interpretation

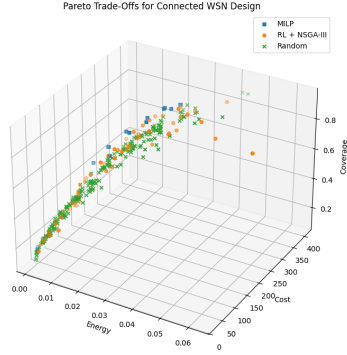
The results establish a clear performance ranking tied to each algorithm’s underlying



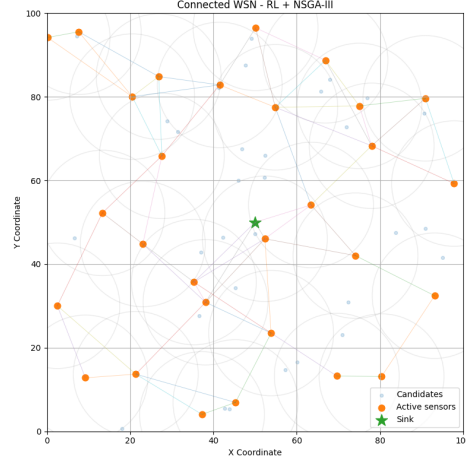
(a) Connected WSN: Coverage vs Energy



(b) Connected WSN: Coverage vs Cost



(c) Pareto Trade-Offs for connected WSN design



(d) Connected WSN coordinates - RL + NSGA-III

Fig. A3: Overall comparison of the four experimental results for seed=7, candidates=60.

logic: MILP forms the strongest benchmark, using systematic optimization to maximize coverage (up to 24 sensors, 270.52 cost) with the lowest energy and hardware requirements. RL + NSGA-III provides a promising, scalable alternative for complex instances, reaching coverage near MILP's level, though its learned configurations require slightly more resources. As for Random Search it lacks optimization guidance, relying on excessive sensor deployment to gain coverage (peaking at 36 sensors, 401.96 cost) and driving up energy and costs.

The Overall Conclusion

MILP forms the ultimate benchmark, generating 14 of the 15 globally non-dominated

solutions. It achieves the highest coverage with the lowest sensor count, energy use, and hardware cost, dominating all Random Search solutions and 14 of 15 RL solutions. RL + NSGA-III serves as an effective, scalable optimizer that nearly matches MILP’s maximum coverage (0.0015 absolute difference) and contributes one globally non-dominated solution. However, its high-coverage options remain more resource-intensive than MILP’s. Random Search performs poorest, yielding zero non-dominated solutions and requiring significantly more sensors, energy, and cost for lower coverage levels.

A.1.4 Sample output simulation results, seed=13 candidates = 60

RL + NSGA-III simulation profile is reported in Table A19.

Table A19: RL + NSGA-III simulation profile

Generation	Best Coverage	epsilon
0	0.8550	0.392
10	0.9038	0.320
20	0.9053	0.262
30	0.9098	0.214
40	0.9246	0.175
50	0.9320	0.143
60	0.9527	0.117
70	0.9645	0.095
80	0.9645	0.078
90	0.9660	0.064
100	0.9630	0.052
110	0.9305	0.042
120	0.9423	0.035
130	0.9645	0.028
140	0.9497	0.023

Table A20: MILP Pareto approximation

Coverage	Energy(J)	Cost	Sensors	Connected
0.9601	0.035663	272.70	25	True
0.9556	0.036300	259.70	24	True
0.9497	0.033058	258.76	24	True
0.9349	0.030958	232.88	22	True
0.9083	0.028371	211.65	20	True
0.8447	0.023025	176.74	17	True
0.8373	0.022693	172.43	17	True
0.7885	0.019701	159.39	15	True
0.7530	0.019166	143.28	14	True
0.6272	0.012189	124.14	11	True
0.3624	0.005670	54.74	6	True
0.3536	0.004990	58.47	6	True
0.2988	0.003913	45.78	5	True
0.1834	0.001659	28.11	3	True
0.1834	0.002319	27.16	3	True

Table A21: RL + NSGA-III Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9349	0.034468	293.75	26	True
0.9246	0.033832	294.12	26	True
0.9246	0.034063	281.40	25	True
0.9216	0.032914	281.80	25	True
0.9201	0.038759	251.81	24	True
0.9201	0.038759	251.81	24	True
0.9186	0.044838	239.46	23	True
0.9172	0.040990	231.39	22	True
0.9127	0.039041	219.44	21	True
0.8994	0.031904	243.29	23	True
0.8950	0.032540	230.29	22	True
0.8935	0.031342	220.62	21	True
0.8935	0.038398	217.94	21	True
0.8831	0.029382	264.26	24	True
0.8757	0.037552	205.57	20	True

Referring to Table A23, for this particular dataset, MILP clearly achieves the largest hypervolume. Its hypervolume is approximately $3.81\times$ that of RL+NSGA-III

Table A22: Random search Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9172	0.066617	343.00	31	True
0.9157	0.052099	386.11	34	True
0.9038	0.036883	295.21	27	True
0.9024	0.038151	283.11	26	True
0.8891	0.048765	281.49	26	True
0.8876	0.032647	252.34	23	True
0.8506	0.044323	242.24	23	True
0.8462	0.034806	241.77	23	True
0.8417	0.034114	247.90	23	True
0.8402	0.031968	251.63	23	True
0.8373	0.031769	249.66	23	True
0.8314	0.030825	254.41	23	True
0.8299	0.027481	232.82	21	True
0.8284	0.027728	228.55	21	True
0.8195	0.026195	207.79	19	True

Table A23: Hypervolume Comparison

Method	Normalized Hypervolume	Relative to RL+NSGA-III
MILP	0.4424	3.81×
Random Search	0.1300	1.12×
RL + NSGA-III	0.1162	1.00×

and $3.40\times$ that of Random Search. Interestingly, Random Search has a slightly larger hypervolume than RL+NSGA-III in this seed, by approximately

$$\frac{0.1300}{0.1162} \approx 1.12,$$

or about 11.9% higher. This does not necessarily mean Random Search is generally superior; it means that under this particular four-objective normalization and reference point, its reported solutions collectively dominate a slightly larger region of the normalized objective space.

Execution Time:

MILP weight sweep :63.129 s, RL + NSGA-III :106.445 s, Random Search : 33.490 s

Unlike several of the other runs where RL + NSGA-III demonstrated the strongest computational advantage, MILP is exceptionally efficient in this particular run. Its execution time of only 63.129 s makes it approximately $1.69\times$ faster than RL + NSGA-III and $2.11\times$ faster than Random Search. Nevertheless, RL + NSGA-III remains computationally preferable to Random Search. This result also indicates that MILP execution time may exhibit greater run-to-run variability, so conclusions regarding overall computational efficiency should preferably be based on the mean execution time across all seeds rather than on this individual run alone.

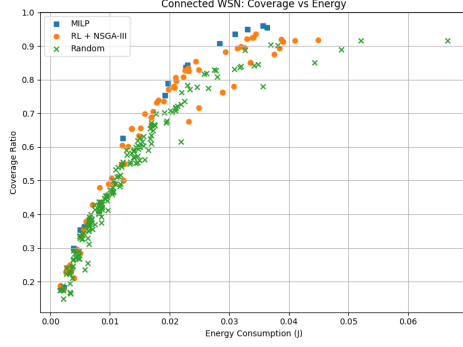
RL operator usage:

Bit Flip : 1235, Add Sensor : 1088

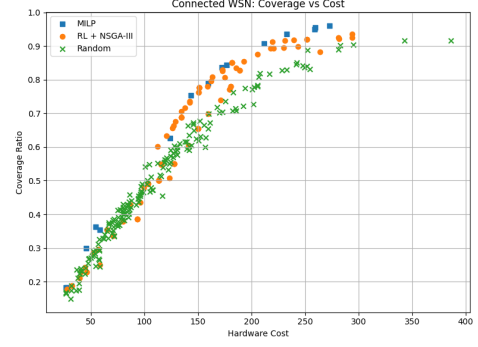
Remove Sensor : 3577, Swap Sensor : 374, Coverage Guided : 8726

Example RL + NSGA-III solution:

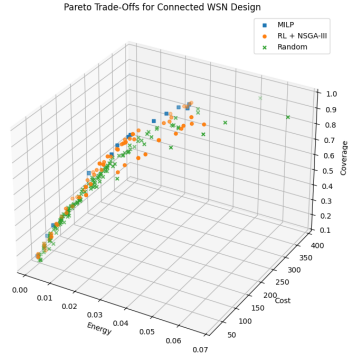
Coverage = 0.9349, Energy = 0.034468 J, Hardware Cost = 293.75, Active Sensors = 26, Connected = True



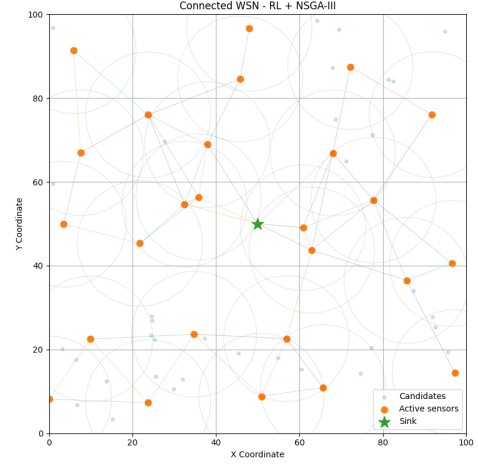
(a) Connected WSN: Coverage vs Energy



(b) Connected WSN: Coverage vs Cost



(c) Pareto Trade-Offs for connected WSN design



(d) Pareto Trade-Offs for connected WSN design

Fig. A4: Overall comparison of the four experimental results for seed=13, candidates=60.

The three tables show a clear hierarchy in solution quality: MILP provides the strongest and most diverse Pareto approximation, RL + NSGA-III provides a competitive but narrower high-coverage front, and Random Search is consistently the

weakest approach. All reported solutions satisfy the connectivity requirement, so the comparison is driven entirely by coverage, energy, cost, and sensor count. For ample comparison details, refer to the comparison Figure A4, Table A20, Table A21, and Table A22.

Overall Comparison

The MILP front has the best coverage capability, reaching 96.01%, compared with

Table A24: Overall Comparison with Seed=13

Criterion	MILP	RL + NSGA-III	Random Search
Maximum coverage	0.9601	0.9349	0.9172
Minimum coverage	0.1834	0.8757	0.8195
Minimum energy (J)	0.001659	0.029382	0.026195
Maximum energy (J)	0.036300	0.044838	0.066617
Minimum cost	27.16	205.57	207.79
Maximum cost	272.70	294.12	386.11
Minimum sensors	3	20	19
Maximum sensors	25	26	34
Maximum-front coverage quality	Best	Second	Third
Trade-off diversity	Very high	Moderate	Moderate

93.49% for RL + NSGA-III and 91.72% for Random Search. More importantly, MILP achieves this higher coverage without requiring the extremely large energy, cost, or sensor counts observed in the Random Search solutions. MILP also provides a much broader Pareto front. Its solutions range from very economical configurations with only 3 sensors, cost ≈ 27 –28, and energy below 0.003 J, up to high-performance configurations with 24–25 sensors and coverage above 95%. This indicates that MILP successfully identifies solutions across the entire design spectrum, from low-resource deployments to maximum-coverage networks. In contrast, RL + NSGA-III concentrates strongly on the high-coverage region, approximately 0.876–0.935. This is not necessarily undesirable: it suggests that the learning-based method has learned to search primarily in the practically attractive high-coverage part of the solution space. However, it does not recover the broad low-cost/low-energy portion of the Pareto front found by MILP.

Overall Interpretation

The results establish the following performance ordering:

MILP > RL + NSGA-III >> Random Search

MILP provides the highest-quality Pareto approximation, achieving the maximum coverage while generally requiring fewer sensors, less energy, and lower cost. It also produces the widest range of trade-off solutions, making it particularly valuable as a reference or benchmark method. Its solutions constitute the dominant part of the combined Pareto front. RL + NSGA-III demonstrates strong optimization capability despite being heuristic. Although it does not match MILP in overall Pareto quality,

it consistently generates high-coverage connected networks and substantially outperforms Random Search in energy, cost, and sensor efficiency. The presence of several globally non-dominated RL solutions also indicates that the method can discover useful alternative trade-offs not directly reproduced by the MILP approximation. Random Search provides the weakest solutions. It requires substantially more sensors, energy, and cost to achieve lower coverage, and all its reported solutions are dominated when compared with MILP. Its primary value is therefore as a baseline demonstrating that the improvements obtained by RL + NSGA-III result from intelligent search rather than chance.

Overall Conclusion

These results support the conclusion that MILP is the preferred approach when high-quality optimization is computationally feasible, while RL + NSGA-III offers a promising approximate alternative for larger instances where exact or MILP-based optimization may become computationally expensive. Random Search serves mainly as a baseline for validating the effectiveness of the proposed intelligent optimization approach.

A.1.5 Sample output simulation results, seed=26 candidates = 60

RL + NSGA-III simulation profile is reported in Table A25.

Table A25: RL + NSGA-III simulation profile

Generation	Best Coverage	epsilon
0	0.8624	0.392
10	0.9290	0.320
20	0.9512	0.262
30	0.9512	0.214
40	0.9512	0.175
50	0.9408	0.143
60	0.9512	0.117
70	0.9512	0.095
80	0.9512	0.078
90	0.9512	0.064
100	0.9512	0.052
110	0.9512	0.042
120	0.9512	0.035
130	0.9512	0.028
140	0.9438	0.023

Table A26: MILP Pareto approximation

Coverage	Energy(J)	Cost	Sensors	Connected
0.9438	0.032862	266.70	24	True
0.9172	0.028534	232.20	21	True
0.9053	0.027570	217.37	20	True
0.8846	0.025436	206.08	19	True
0.8506	0.023305	192.46	18	True
0.7870	0.018932	175.99	16	True
0.7293	0.016731	148.87	14	True
0.7130	0.017136	137.25	13	True
0.7056	0.015083	151.52	13	True
0.2530	0.003318	36.97	4	True
0.2426	0.002746	36.88	4	True
0.1923	0.002155	26.86	3	True
0.1820	0.001537	27.56	3	True

Referring to Table A29, MILP achieves approximately $3.10\times$ the hypervolume of RL+NSGA-III and $5.88\times$ that of Random Search. RL+NSGA-III achieves approximately $1.90\times$ the hypervolume of Random Search, corresponding to about an 89.7% larger dominated objective-space volume.

Execution Time:

MILP weight sweep :132.002 s, RL + NSGA-III :107.321 s, Random Search :131.957 s

The results demonstrate a clear computational advantage for RL + NSGA-III, which achieves the lowest execution time of 107.321 s. It reduces computation time

Table A27: RL + NSGA-III Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9438	0.043530	354.13	31	True
0.9112	0.036421	306.02	27	True
0.9024	0.030734	267.46	24	True
0.9024	0.030734	267.46	24	True
0.8920	0.030682	245.90	22	True
0.8861	0.030441	244.90	22	True
0.8861	0.029105	257.08	23	True
0.8846	0.028162	248.50	22	True
0.8772	0.027607	232.37	21	True
0.8698	0.027705	224.10	20	True
0.8683	0.027361	214.84	19	True
0.8683	0.027361	214.84	19	True
0.8683	0.025578	223.15	20	True
0.8683	0.025578	223.15	20	True
0.8565	0.025182	210.64	19	True

Table A28: Random search Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.8994	0.050660	376.90	33	True
0.8964	0.036007	338.23	29	True
0.8817	0.037152	325.19	27	True
0.8817	0.041609	301.31	26	True
0.8802	0.032013	306.28	26	True
0.8743	0.034620	288.41	25	True
0.8728	0.032438	297.28	26	True
0.8595	0.041295	262.78	24	True
0.8550	0.031054	292.45	25	True
0.8550	0.035584	245.76	22	True
0.8476	0.034566	273.45	23	True
0.8476	0.034296	287.97	25	True
0.8476	0.032962	290.00	25	True
0.8447	0.029894	249.50	22	True
0.8417	0.031971	246.08	21	True

Table A29: Hypervolume Comparison

Method	Normalized Hypervolume	Relative to Random Search
MILP	0.3293	5.88×
RL + NSGA-III**	0.1063	1.90×
Random Search**	0.0560	1.00×

by approximately 18.7% compared with both MILP and Random Search. In contrast, MILP and Random Search require virtually identical execution times of approximately 132 s. Therefore, for this particular run, RL + NSGA-III provides the best computational efficiency, while MILP and Random Search are essentially equivalent from a runtime perspective.

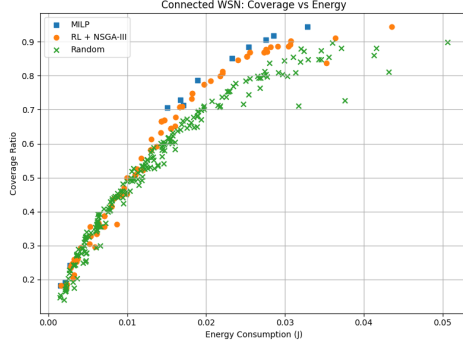
RL operator usage:

Bit Flip : 1856, Add Sensor : 1032

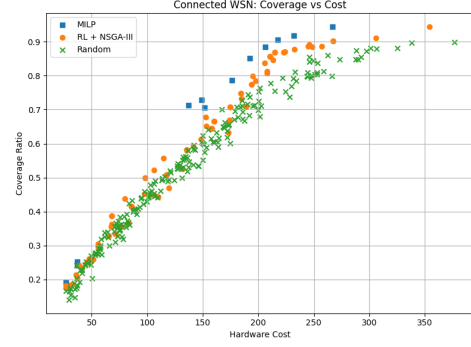
Remove Sensor : 2282, Swap Sensor : 413, Coverage Guided : 9417

Example RL + NSGA-III Solution:

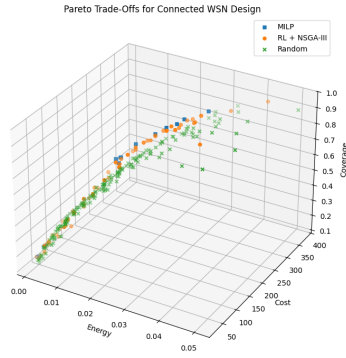
Coverage = 0.9438, Energy = 0.043530 J, Hardware Cost = 354.13, Active Sensors = 31, Connected = True



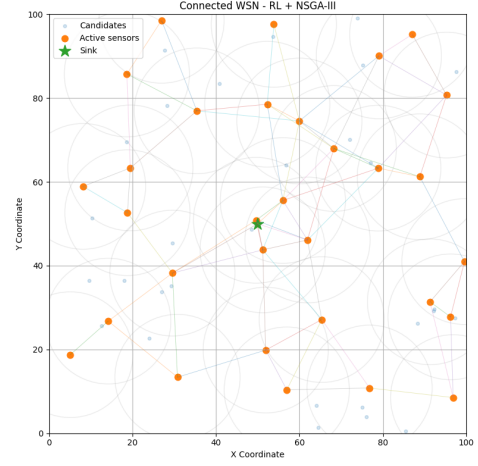
(a) Connected WSN: Coverage vs Energy



(b) Connected WSN: Coverage vs Cost



(c) Pareto Trade-Offs for connected WSN



(d) Connected WSN coordinates - RL + NSGA-III

Fig. A5: Overall comparison of the four experimental results for seed=26, candidates=60.

The three Pareto sets show a clear hierarchy in solution quality. MILP provides the strongest overall trade-offs, RL + NSGA-III provides good high-coverage solutions but at higher resource consumption, while Random Search is substantially less efficient. Importantly, every solution in all three methods satisfies the connectivity constraint (Connected = True), so the comparison can focus directly on coverage,

energy, cost, and sensor count. For ample comparison details, refer to the comparison Figure A5, Table A26, Table A27, and Table A28.

Overall Comparison

Table A30: Overall Comparison with Seed=26

Criterion	MILP	RL + NSGA-III	Random Search
Maximum coverage	0.9438	0.9438	0.8994
Minimum coverage represented	0.1820	0.8565	0.8417
Energy range (J)	0.001537–0.032862	0.025182–0.043530	0.029894–0.050660
Cost range	26.86–266.70	210.64–354.13	245.76–376.90
Sensor range	3–24	19–31	21–33
Connectivity	100%	100%	100%
Unique listed solutions	13/13	12/15	15/15

The three methods also differ in the shape and usefulness of their fronts. MILP produces the widest and most informative Pareto front. It captures the full transition from very inexpensive low-coverage networks to high-coverage networks. RL + NSGA-III concentrates predominantly on the high-quality region. This can actually be desirable when the primary design goal is high coverage, although it does not provide the same wide range of low-resource alternatives as MILP. Random Search is concentrated within a relatively narrow coverage interval of 0.8417–0.8994, but its energy and cost remain high. Thus, even though it generates multiple nondominated solutions internally, these solutions are weak when compared globally against the other methods.

Overall Interpretation

The results establish the following performance ranking:
MILP > RL + NSGA-III >> Random Search
MILP serves as the strongest benchmark because it achieves equal or higher coverage with substantially fewer sensors, lower energy consumption, and lower hardware cost. Its Pareto approximation is also much broader, providing decision makers with a richer selection of trade-offs. RL + NSGA-III nevertheless shows strong optimization capability. It successfully reaches the same maximum coverage as MILP and clearly outperforms Random Search. Its main disadvantage is that it requires additional sensors, energy, and cost to achieve comparable coverage. For a heuristic approach, however, its ability to approach the MILP coverage level is encouraging, especially if the method is intended for larger instances where exact MILP optimization may become computationally expensive. Random Search provides an appropriate baseline but performs poorly. Its inability to exceed 89.94% coverage despite using up to 33 sensors and the highest energy and cost levels demonstrates the value of directed optimization.

Overall Conclusion

These results therefore support the conclusion that RL + NSGA-III provides a meaningful approximation to the high-coverage region of the MILP Pareto front and substantially improves upon uninformed Random Search, while MILP remains the reference method in terms of Pareto efficiency and resource utilization.

A.2 Sample Solution where the Number of Candidates = 60

A.2.1 Sample output simulation results, seed=1 candidates = 150

RL + NSGA-III simulation profile is reported in Table A31.

Table A31: RL + NSGA-III simulation profile

Generation	Best Coverage	epsilon
0	0.9660	0.392
10	0.9867	0.320
20	0.9970	0.262
30	1.0000	0.214
40	0.9941	0.175
50	0.9941	0.143
60	1.0000	0.117
70	0.9985	0.095
80	0.9882	0.078
90	0.9911	0.064
100	0.9911	0.052
110	0.9911	0.042
120	0.9911	0.035
130	0.9911	0.028
140	0.9719	0.023

Table A32: MILP Pareto approximation

Coverage	Energy(J)	Cost	Sensors	Connected
1.0000	0.033134	259.10	25	True
0.9985	0.032017	256.83	25	True
0.9985	0.032477	252.04	25	True
0.9970	0.031815	249.73	25	True
0.9911	0.029588	247.26	24	True
0.9911	0.030716	236.08	23	True
0.9719	0.026409	226.23	21	True
0.9675	0.027038	204.86	20	True
0.9645	0.027243	199.87	20	True
0.9527	0.025024	196.31	19	True
0.9482	0.023564	206.03	20	True
0.8950	0.019399	181.89	18	True
0.8891	0.022251	159.46	17	True
0.8669	0.018738	160.35	17	True
0.5843	0.009660	86.25	10	True

Table A33: RL + NSGA-III Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9675	0.045899	329.59	28	True
0.9675	0.054735	327.94	28	True
0.9615	0.043418	327.64	28	True
0.9571	0.065413	323.22	28	True
0.9556	0.063900	314.44	27	True
0.9556	0.064288	310.64	27	True
0.9512	0.029334	251.90	23	True
0.9497	0.028357	269.02	23	True
0.9467	0.028478	240.75	23	True
0.9423	0.027511	274.96	24	True
0.9423	0.027511	274.96	24	True
0.9408	0.027528	261.84	23	True
0.9393	0.025889	265.41	23	True
0.9379	0.026238	256.70	22	True
0.9172	0.026222	255.77	23	True

Referring to Table A35, MILP achieves approximately $1.66\times$ the hypervolume of RL+NSGA-III and $4.88\times$ that of Random Search. RL+NSGA-III achieves approximately $2.95\times$ the hypervolume of Random Search.

Table A34: Random search Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
1.0000	0.073765	641.65	54	True
0.9985	0.066750	590.95	52	True
0.9970	0.068626	576.54	51	True
0.9956	0.067964	577.90	49	True
0.9941	0.064201	502.70	44	True
0.9926	0.060114	500.11	43	True
0.9837	0.054015	471.35	41	True
0.9808	0.059206	453.76	40	True
0.9793	0.057363	456.72	39	True
0.9778	0.048271	436.07	38	True
0.9763	0.052360	433.01	38	True
0.9719	0.042112	376.89	33	True
0.9586	0.045516	376.15	35	True
0.9512	0.047099	364.95	32	True
0.9453	0.042072	354.99	33	True

Table A35: Hypervolume Comparison

Method	Normalized Hypervolume	Relative to Random Search
MILP	0.5794	4.88×
RL + NSGA-III	0.3498	2.95×
Random Search	0.1187	1.00×

Execution Time:

The most striking result is computational performance:

MILP: 19,412.029 s $\approx$ 5.39 hours

RL + NSGA-III: 222.646 s $\approx$ 3.71 minutes

Random Search: 508.102 s $\approx$ 8.47 minutes

RL + NSGA-III is therefore approximately 87.2 times faster than MILP, corresponding to a 98.85% reduction in execution time. It is also approximately 2.28 times faster than Random Search, corresponding to a 56.2% reduction in execution time. Even Random Search is approximately 38.2 times faster than MILP. Consequently, the superior Pareto quality of MILP comes at a very substantial computational cost.

RL operator usage:

Bit Flip : 1164, Add Sensor : 988

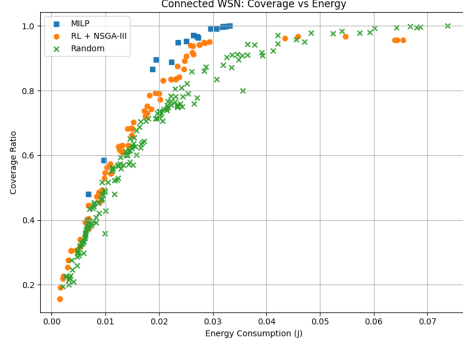
Remove Sensor : 4324, Swap Sensor : 352

Coverage Guided : 8172

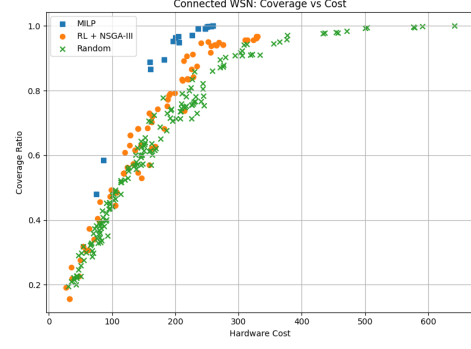
Example RL + NSGA-III Solution:

Coverage = 0.9675, Energy = 0.045899 J, Hardware Cost = 329.59, Active Sensors = 28, Connected = True

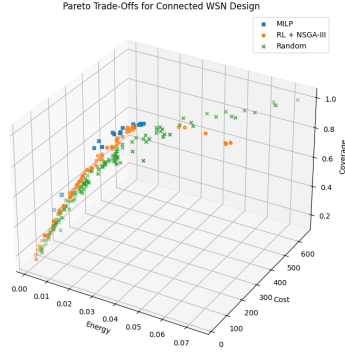
The three tables show a clear hierarchy in solution quality: The MILP weight-sweep provides the strongest overall Pareto approximation, reaching complete coverage while consistently requiring substantially fewer sensors, less energy, and lower hardware cost. In the reported experiment, all RL + NSGA-III and Random Search solutions are



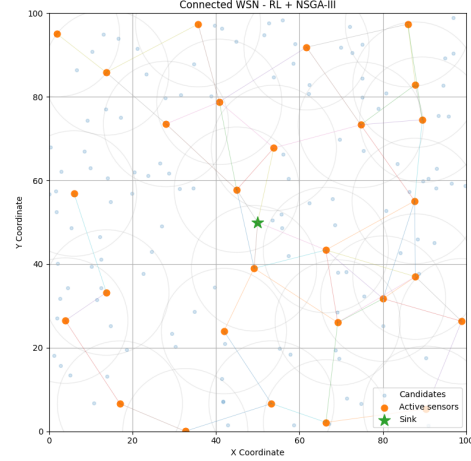
(a) Connected WSN: Coverage vs Cost



(b) Connected WSN: Coverage vs Cost



(c) Pareto Trade-Offs for connected WSN design



(d) Connected WSN coordinates - RL + NSGA-III

Fig. A6: Overall comparison of the four experimental results for seed=1, candidates=150.

dominated by at least one MILP solution, confirming the superior resource efficiency of the exact optimization framework. Nevertheless, this advantage is obtained at an extremely high computational cost: MILP requires approximately hours, compared with to few minutes for RL + NSGA-III. For ample comparison details, refer to the comparison Figure A6, Table A32, Table A33, and Table A34.

Overall comparison

Overall, the results demonstrate a clear trade-off between solution quality and computational efficiency. The MILP weight-sweep produces the most resource-efficient

Table A36: Overall Comparison with Seed=1-150

Criterion	MILP	RL + NSGA-III	Random Search
Maximum Coverage	1.0000 (100%)	0.9675 (96.75%)	1.0000 (100%)
Coverage Quality	Best	High	High, but resource-intensive
Energy Efficiency	Best	Moderate to good	Worst
Hardware Cost Efficiency	Best	Moderate to good	Worst
Sensor Efficiency	Best	Moderate	Worst
Pareto Front Quality	Best / strongest	Good, but dominated by MILP in this run	Weakest
Range of Trade-offs	Broadest	Narrower, concentrated in high coverage	Mainly high-resource solutions
Connectivity	Guaranteed in all reported solutions	Guaranteed in all reported solutions	Guaranteed in all reported solutions
Execution Time	19,412.029 s	222.646 s	508.102 s
Computational Speed	Slowest	Fastest	Intermediate
Speed vs MILP	Baseline	87.2× faster	≈ 38.2× faster
Scalability Potential	Limited by computational cost	Best	Moderate
Overall Strength	Best solution/resource efficiency	Best balance of quality and speed	Simple baseline but inefficient
Overall Ranking	1st in solution quality	1st in computational efficiency	3rd overall

Pareto solutions, achieving up to 100% coverage with substantially lower energy consumption, hardware cost, and sensor count than both RL + NSGA-III and Random Search. However, this superior solution quality comes at a very high computational cost, requiring approximately 19,412 s. In contrast, RL + NSGA-III completes the optimization in only 222.646 s, making it approximately 87 times faster than MILP and about 2.28 times faster than Random Search. Although RL + NSGA-III reaches a slightly lower maximum coverage of 96.75%, it provides considerably more efficient solutions than Random Search in terms of energy, cost, and number of deployed sensors. Random Search is able to achieve 100% coverage, but only through substantial overdeployment of sensors and correspondingly high energy consumption and cost. Therefore, MILP is the preferred approach when solution quality and resource efficiency are prioritized, whereas RL + NSGA-III provides the most attractive balance between high-quality feasible solutions and computational scalability. Random Search remains the weakest approach overall because its high-coverage solutions are obtained at significantly greater resource expense.

Overall interpretation

The results highlight a clear quality-versus-computation trade-off: MILP delivers the best Pareto approximation, achieving 100% coverage with the lowest sensor count, energy usage, and hardware cost. However, all solutions dominate RL + NSGA-III and Random Search at a steep runtime of ≈ 5.39 hours. RL + NSGA-III: serves as a practical alternative for large or time-sensitive WSN designs. It runs ≈ 87 times faster than MILP (3.71 minutes) and yields far cheaper solutions than Random

Search, though its maximum coverage caps at 96.75%. Random Search reaches 100% coverage, but relies on over-deploying sensors, causing excessive energy and hardware costs. The recommendation is to use MILP when absolute resource efficiency and coverage are required; an to use RL + NSGA-III when computational speed and scalability take priority.

Overall conclusion

Overall, MILP provides the most resource-efficient solutions but at a very high computational cost. RL + NSGA-III offers the best balance between solution quality and scalability, running approximately 87 times faster than MILP and more than twice as fast as Random Search while maintaining high-quality feasible solutions. Random Search remains inferior in resource utilization despite achieving high coverage.

A.2.2 Sample output simulation results, seed=3 candidates = 150

RL + NSGA-III simulation profile is reported in Table A37.

Table A37: RL + NSGA-III simulation profile

Generation	Best Coverage	epsilon
0	0.9630	0.392
10	0.9822	0.320
20	0.9911	0.262
30	0.9970	0.214
40	1.0000	0.175
50	1.0000	0.143
60	0.9941	0.117
70	1.0000	0.095
80	0.9926	0.078
90	0.9852	0.064
100	0.9941	0.052
110	1.0000	0.042
120	1.0000	0.035
130	0.9896	0.028
140	0.9852	0.023

Table A38: MILP Pareto approximation

Coverage	Energy(J)	Cost	Sensors	Connected
0.9985	0.031940	280.89	26	True
0.9970	0.030775	275.66	25	True
0.9956	0.030819	262.65	24	True
0.9896	0.028220	253.17	23	True
0.9896	0.027846	253.97	23	True
0.9808	0.026902	233.72	22	True
0.9749	0.025629	235.02	21	True
0.9719	0.025985	221.05	21	True
0.9275	0.021127	199.79	19	True
0.9231	0.020969	196.52	19	True
0.8935	0.019298	189.09	18	True
0.8831	0.019837	175.12	17	True
0.4172	0.006858	61.15	7	True
0.3180	0.003067	51.65	5	True

Table A39: RL + NSGA-III Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9852	0.038960	305.57	26	True
0.9630	0.028507	269.52	23	True
0.9630	0.029318	257.27	23	True
0.9615	0.028285	285.90	25	True
0.9586	0.027268	265.78	23	True
0.9541	0.031259	234.26	22	True
0.9541	0.026679	272.12	23	True
0.9438	0.025549	253.88	22	True
0.9408	0.031155	226.22	21	True
0.9379	0.030556	236.99	22	True
0.9364	0.026151	238.90	22	True
0.9364	0.026151	238.90	22	True
0.9364	0.026151	238.90	22	True
0.9231	0.026050	238.01	21	True
0.8994	0.023781	224.92	20	True

Table A40: Random search Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9956	0.070291	681.08	57	True
0.9956	0.077552	665.08	58	True
0.9926	0.064651	565.98	48	True
0.9896	0.059022	500.93	45	True
0.9837	0.057350	564.65	49	True
0.9822	0.056470	559.06	50	True
0.9808	0.059278	472.71	41	True
0.9793	0.054799	542.77	45	True
0.9793	0.056415	505.47	44	True
0.9763	0.054946	486.13	42	True
0.9749	0.072375	472.65	41	True
0.9749	0.053745	525.32	44	True
0.9719	0.053820	456.40	39	True
0.9704	0.051564	486.01	41	True
0.9689	0.052035	427.64	38	True

Table A41: Hypervolume Comparison

Method	Normalized Hypervolume	Relative to Random Search
MILP	0.5273	10.29×
RL + NSGA-III	0.3582	6.99×
Random Search	0.0512	1.00×

Referring to Table A41, MILP achieves approximately $1.47\times$ the hypervolume of RL+NSGA-III and $10.29\times$ that of Random Search. RL+NSGA-III achieves approximately $6.99\times$ the hypervolume of Random Search.

Execution Time:

MILP weight sweep : 28674.932 s $\approx$ 7.97 hours

RL + NSGA-III : 241.289 s $\approx$ 4.02 minutes

Random Search : 537.668 s $\approx$ 8.96 minutes

RL + NSGA-III is therefore: approximately $118.8\times$ faster than MILP;
equivalent to a 99.16% reduction in execution time relative to MILP;
approximately $2.23\times$ faster than Random Search;
approximately 55.1% faster than Random Search.

MILP is approximately $53.3\times$ slower than Random Search.

This computational difference is particularly significant. Although MILP clearly produces the strongest Pareto solutions, obtaining those solutions requires almost eight hours, compared with only about four minutes for RL + NSGA-III.

RL operator usage:

Bit Flip : 1244, Add Sensor : 518

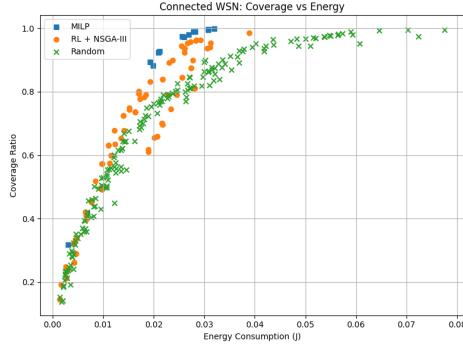
Remove Sensor : 3403, Swap Sensor : 497

Coverage Guided : 9338

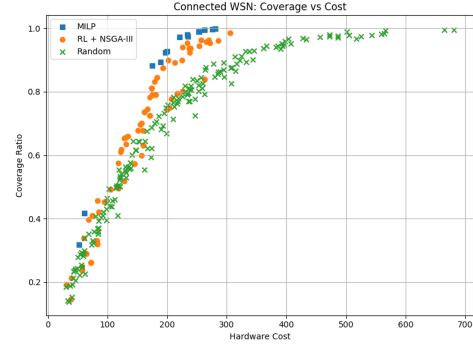
Example RL + NSGA-III Solution

Coverage = 0.9852, Energy = 0.038960 J, Hardware Cost = 305.57, Active Sensors

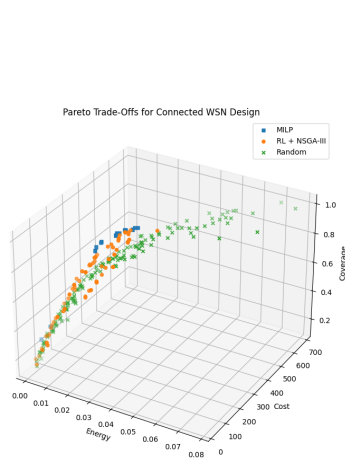
= 26, Connected = True



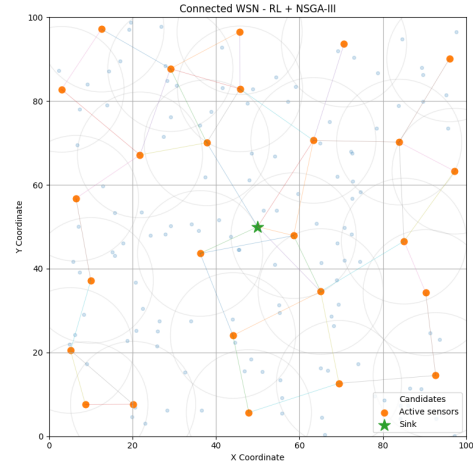
(a) Connected WSN: Coverage vs Cost



(b) Connected WSN: Coverage vs Cost



(c) Pareto Trade-Offs for connected WSN design



(d) Connected WSN coordinates - RL + NSGA-III

Fig. A7: Overall comparison of the four experimental results for seed=3, candidates=150.

The three tables show a clear hierarchy in solution quality: The MILP weight-sweep provides the strongest overall Pareto approximation, reaching complete coverage while consistently requiring substantially fewer sensors, less energy, and lower hardware cost. In the reported experiment, all RL + NSGA-III and Random Search solutions are dominated by at least one MILP solution, confirming the superior resource efficiency of the exact optimization framework. Nevertheless, this advantage is obtained at an extremely high computational cost: MILP requires approximately

hours, compared with to few minutes for RL + NSGA-III. For ample comparison details, refer to the comparison Figure A7, Table A38, Table A39, and Table A40.

Overall comparison

The three tables show a clear separation between the methods in terms of Pareto quality, resource efficiency, and computational cost. All reported solutions satisfy the connectivity constraint, so the comparison can focus directly on coverage, energy, hardware cost, sensor count, and runtime.

Table A42: Overall Comparison with Seed=3-150

Criterion	MILP	RL + NSGA-III	Random Search
Maximum coverage	0.9985	0.9852	0.9956
Energy at maximum coverage	0.031940 J	0.038960 J	0.070291 J *
Cost at maximum coverage	280.89	305.57	681.08*
Sensors at maximum coverage	26	26	57*
Coverage range	0.3180–0.9985	0.8994–0.9852	0.9689–0.9956
Execution time	28,674.932 s	241.289 s	537.668 s
Main advantage	Pareto/resource efficiency	Computational speed	Simple baseline
Overall ranking	Best solution quality	Best speed/quality compromise	Weakest resource efficiency

* using the first Random Search solution at coverage 0.9956.

Overall Interpretation

Overall, the results reveal a clear trade-off between solution quality and computational efficiency among the three approaches. MILP provides the strongest Pareto approximation, achieving the highest coverage while generally requiring fewer sensors, lower energy consumption, and lower hardware cost than the competing methods at comparable coverage levels. However, this superior resource efficiency is obtained at a very high computational cost, with an execution time of approximately 28,675 seconds. In contrast, RL + NSGA-III achieves slightly lower Pareto efficiency but requires only 241 seconds, making it approximately 119 times faster than MILP. It also substantially outperforms Random Search in resource utilization while being more than twice as fast. Random Search can attain very high coverage, but it does so by deploying considerably more sensors and consequently incurs much higher energy consumption and hardware cost. Therefore, MILP is the preferred approach when solution quality and resource optimality are prioritized, whereas RL + NSGA-III provides the most attractive overall compromise between high-quality WSN designs and computational scalability. Random Search remains primarily useful as a baseline due to its comparatively inefficient use of network resources.

Overall Conclusion

The results reveal a clear trade-off between Pareto efficiency and computational scalability. MILP delivers the most resource-efficient solutions, dominating all Random

Search solutions and 14 of the 15 RL + NSGA-III solutions, but at a very high computational cost of about 7.97 hours. In contrast, RL + NSGA-III completes in only 4.02 minutes, making it approximately 119 times faster while still producing connected, high-coverage solutions that substantially outperform Random Search. Thus, MILP is preferable when solution optimality is prioritized, whereas RL + NSGA-III offers the best compromise for large-scale or time-sensitive WSN design.

A.2.3 Sample output simulation results, seed=7 candidates = 150

RL + NSGA-III simulation profile is reported in Table A43:

Table A43: RL + NSGA-III simulation profile

Generation	Best Coverage	epsilon
0	0.9630	0.392
10	0.9956	0.320
20	0.9970	0.262
30	1.0000	0.214
40	1.0000	0.175
50	1.0000	0.143
60	0.9926	0.117
70	1.0000	0.095
80	1.0000	0.078
90	0.9926	0.064
100	0.9882	0.052
110	1.0000	0.042
120	1.0000	0.035
130	0.9941	0.028
140	0.9941	0.023

Table A44: MILP Pareto approximation

Coverage	Energy(J)	Cost	Sensors	Connected
0.9985	0.031311	281.92	24	True
0.9970	0.030899	271.51	24	True
0.9970	0.032891	260.08	24	True
0.9911	0.029879	254.14	23	True
0.9896	0.028006	264.51	23	True
0.9793	0.028485	232.41	22	True
0.9704	0.024348	244.57	21	True
0.9704	0.025075	236.39	21	True
0.9660	0.024406	233.69	21	True
0.9615	0.023620	238.08	21	True
0.9541	0.025914	208.62	20	True
0.8876	0.022486	170.40	17	True
0.8817	0.019015	179.75	17	True
0.3683	0.005976	53.72	6	True
0.3521	0.003978	69.92	6	True

Table A45: RL + NSGA-III Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
1.0000	0.045487	396.73	33	True
0.9867	0.031552	311.92	25	True
0.9808	0.030359	328.21	26	True
0.9793	0.033724	269.38	24	True
0.9778	0.029439	313.26	25	True
0.9749	0.031013	255.14	23	True
0.9601	0.027850	299.43	24	True
0.9601	0.027850	299.43	24	True
0.9541	0.028993	234.32	21	True
0.9512	0.026960	303.14	24	True
0.9438	0.028976	236.05	21	True
0.9349	0.025844	269.73	22	True
0.9275	0.027305	224.22	20	True
0.9098	0.024089	257.08	21	True
0.9098	0.030884	212.83	19	True

Referring to Table A47, MILP therefore achieves approximately $1.21\times$ the hypervolume of RL+NSGA-III and $3.39\times$ that of Random Search. RL+NSGA-III achieves approximately $2.81\times$ the hypervolume of Random Search, corresponding to about a 180.6% larger dominated objective-space region.

Table A46: Random search Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9985	0.114994	749.27	63	True
0.9970	0.072062	604.19	53	True
0.9956	0.064310	549.83	48	True
0.9941	0.062118	584.45	50	True
0.9911	0.064667	501.48	44	True
0.9911	0.059419	550.89	48	True
0.9867	0.057545	551.47	45	True
0.9852	0.066154	476.62	42	True
0.9852	0.058131	495.77	41	True
0.9837	0.057263	530.91	44	True
0.9822	0.054799	467.68	39	True
0.9808	0.049854	445.01	38	True
0.9749	0.046537	405.87	35	True
0.9615	0.041097	402.31	34	True
0.9586	0.048840	395.43	34	True

Table A47: Hypervolume Comparison

Method	Normalized Hypervolume	Relative to Random Search
MILP	0.5672	3.39×
RL + NSGA-III	0.4691	2.81×
Random Search	0.1672	1.00×

Execution Time

MILP weight sweep :17353.829s, RL + NSGA-III :226.98s, Random Search :545.484s

RL + NSGA-III reduces execution time relative to MILP by approximately 98.69%, making it about 76.45 times faster. It is also about 2.40 times faster than Random Search, corresponding to approximately a 58.4% reduction in runtime. Random Search itself is about 31.8 times faster than the MILP weight sweep, but this speed comes with substantially poorer solution quality.

RL operator usage:

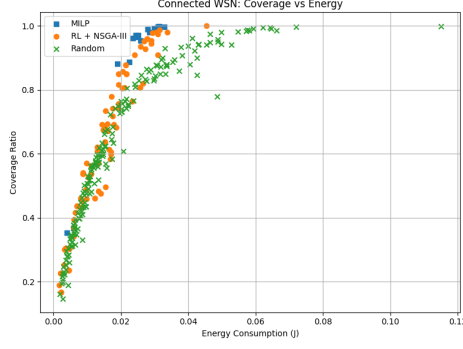
Bit Flip : 988, Add Sensor : 628

Remove Sensor : 2796, Swap Sensor : 426

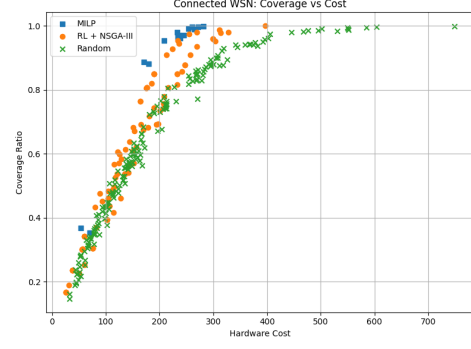
Coverage Guided : 10162

Example RL + NSGA-III Solution Coverage = 1.0000, Energy = 0.045487 J, Hardware Cost = 396.73, Active Sensors = 33, Connected = True

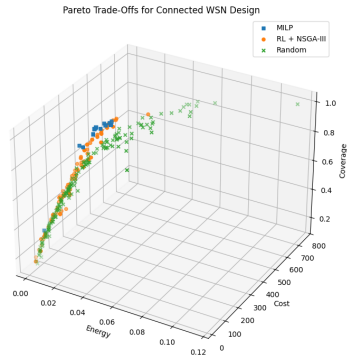
The three tables show a clear hierarchy in solution quality: The MILP weight-sweep provides the strongest overall Pareto approximation, reaching complete coverage while consistently requiring substantially fewer sensors, less energy, and lower hardware cost. In the reported experiment, all RL + NSGA-III and Random Search solutions are dominated by at least one MILP solution, confirming the superior resource efficiency of the exact optimization framework. Nevertheless, this advantage is obtained at an extremely high computational cost: MILP requires approximately hours, compared with to few minutes for RL + NSGA-III. For ample comparison



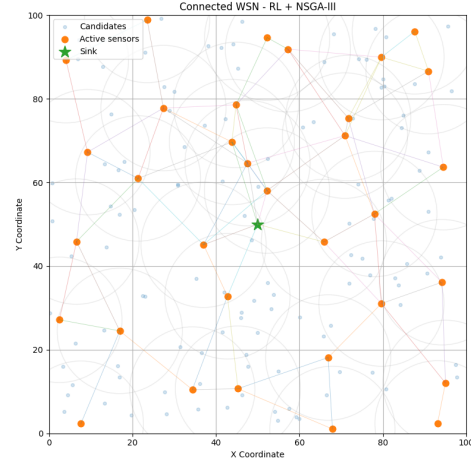
(a) Connected WSN: Coverage vs Cost



(b) Connected WSN: Coverage vs Cost



(c) Pareto Trade-Offs for connected WSN design



(d) Connected WSN coordinates - RL + NSGA-III

Fig. A8: Overall comparison of the four experimental results for seed=7, candidates=150.

details, refer to the comparison Figure A8, Table A44, Table A45, and Table A46.

Overall Comparison

The three tables show a very clear hierarchy. MILP provides the strongest overall Pareto efficiency, RL + NSGA-III achieves the absolute highest coverage and the best computational efficiency, while Random Search is substantially inferior in resource utilization.

Overall Interpretation

The results demonstrate a clear trade-off between Pareto efficiency and computational efficiency among the three approaches. MILP provides the strongest overall solution

Table A48: Overall Comparison with Seed=7-150

Criterion	MILP	RL + NSGA-III	Random Search	Best
Maximum coverage	99.85%	100.00%	99.85%	RL + NSGA-III
Energy at maximum coverage	0.031311 J	0.045487 J	0.114994 J	MILP
Cost at maximum coverage	281.92	396.73	749.27	MILP
Sensors at maximum coverage	24	33	63	MILP
Minimum sensors represented	6	19	34	MILP
Execution time	17,353.829 s	226.984 s	545.484 s	RL + NSGA-III
Connectivity	100%	100%	100%	Equal

quality, generating a broad and highly resource-efficient Pareto approximation. It reaches 99.85% coverage with only 24 sensors, compared with 63 sensors required by Random Search at exactly the same coverage, corresponding to reductions of approximately 72.8% in energy consumption, 62.4% in hardware cost, and 61.9% in sensor count. MILP is also generally more resource-efficient than RL + NSGA-III at comparable coverage levels. Indeed, when the three primary objectives of coverage, energy, and cost are considered, 14 of the 15 reported RL + NSGA-III solutions are dominated by at least one MILP solution, while all Random Search solutions are dominated by both optimized approaches. Nevertheless, RL + NSGA-III presents two important advantages. First, it is the only method to achieve complete 100% coverage, although this requires additional energy, cost, and sensors relative to MILP’s near-complete 99.85% solution. Second, and more importantly from a scalability perspective, RL + NSGA-III requires only 226.984 s compared with 17,353.829 s for the MILP weight sweep, making it approximately 76.45 times faster and reducing computation time by about 98.69%. It is also approximately 2.40 times faster than Random Search. Random Search, despite achieving high coverage, relies on substantial sensor over-deployment and consequently exhibits considerably higher energy consumption and hardware cost. Overall, MILP should be regarded as the benchmark for Pareto and resource efficiency, whereas RL + NSGA-III provides the most attractive balance between solution quality, maximum achievable coverage, and computational efficiency. The results therefore support RL + NSGA-III as a scalable alternative to exact MILP optimization, particularly when computational time is critical, while Random Search serves primarily as a baseline and is clearly inferior in terms of deployment efficiency.

Overall Conclusion

Overall, MILP produces the most resource-efficient Pareto solutions, achieving near-complete coverage with substantially lower energy, cost, and sensor count. RL + NSGA-III reaches the highest coverage of 100% and is approximately 76 times faster than MILP, offering the best compromise between solution quality and computational efficiency. Random Search is consistently less efficient and mainly serves as a baseline.

A.2.4 Sample output simulation results, seed=13 candidates = 150

RL + NSGA-III simulation profile is reported in Table A49:

Table A49: RL + NSGA-III simulation profile

Generation	Best Coverage	epsilon
0	0.9497	0.392
10	0.9911	0.320
20	0.9985	0.262
30	0.9985	0.214
40	0.9985	0.175
50	0.9985	0.143
60	0.9985	0.117
70	0.9985	0.095
80	0.9867	0.078
90	0.9985	0.064
100	0.9985	0.052
110	0.9970	0.042
120	0.9985	0.035
130	0.9985	0.028
140	0.9985	0.023

Table A50: MILP Pareto approximation

Coverage	Energy(J)	Cost	Sensors	Connected
0.9985	0.032467	273.63	25	True
0.9970	0.030982	268.10	25	True
0.9970	0.032422	256.56	25	True
0.9970	0.031408	263.54	25	True
0.9941	0.031726	248.97	24	True
0.9867	0.028160	249.47	23	True
0.9837	0.028295	241.89	23	True
0.9749	0.025561	239.00	22	True
0.9645	0.023890	233.28	21	True
0.9630	0.023990	229.22	21	True
0.9556	0.022901	227.49	20	True
0.9453	0.024964	203.06	20	True
0.8772	0.017609	200.88	17	True
0.8624	0.018227	167.81	16	True
0.8565	0.019409	158.02	16	True

Table A51: RL + NSGA-III Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9911	0.035584	340.68	29	True
0.9837	0.033401	326.07	28	True
0.9808	0.042704	300.57	26	True
0.9808	0.047972	268.42	24	True
0.9719	0.032773	311.79	27	True
0.9675	0.045178	265.72	24	True
0.9660	0.040907	286.10	25	True
0.9630	0.042347	257.24	23	True
0.9527	0.044830	251.25	23	True
0.9527	0.044830	251.25	23	True
0.9482	0.043129	256.24	23	True
0.9482	0.033817	300.77	25	True
0.9467	0.036176	280.31	25	True
0.9364	0.035408	265.72	24	True
0.9349	0.029154	287.77	24	True

Referring to Table A53, MILP achieves approximately $1.47\times$ the hypervolume of RL+NSGA-III and $6.86\times$ that of Random Search. RL+NSGA-III achieves approximately $4.66\times$ the hypervolume of Random Search.

Execution Time

MILP weight sweep : 8243.659 s $\approx$ 2.29 hours

RL + NSGA-III : 180.056 s , Random Search : 495.542 s

RL + NSGA-III is approximately $45.8\times$ faster than MILP, corresponding to a 97.8% reduction in execution time. It is also approximately $2.75\times$ faster than Random Search, or about 63.7% faster. Random Search itself is approximately $16.6\times$ faster than MILP, but this speed comes with substantially poorer resource efficiency.

Table A52: Random search Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9970	0.078400	742.32	63	True
0.9941	0.080928	724.60	61	True
0.9926	0.072363	612.21	53	True
0.9911	0.068680	595.34	52	True
0.9867	0.066048	523.17	46	True
0.9837	0.059973	513.38	45	True
0.9734	0.057804	473.19	40	True
0.9734	0.057091	503.80	44	True
0.9675	0.055445	506.10	44	True
0.9660	0.053382	491.22	42	True
0.9645	0.056132	487.81	42	True
0.9645	0.057599	467.49	40	True
0.9630	0.052781	449.91	40	True
0.9601	0.055878	437.37	38	True
0.9571	0.050859	445.85	38	True

Table A53: Hypervolume Comparison

Method	Normalized Hypervolume	Relative to Random Search
MILP	0.7469	6.86×
RL + NSGA-III	0.5078	4.66×
Random Search	0.1089	1.00×

RL operator usage: Bit Flip : 1391, Add Sensor : 1439

Remove Sensor : 3211, Swap Sensor : 545

Coverage Guided : 8414

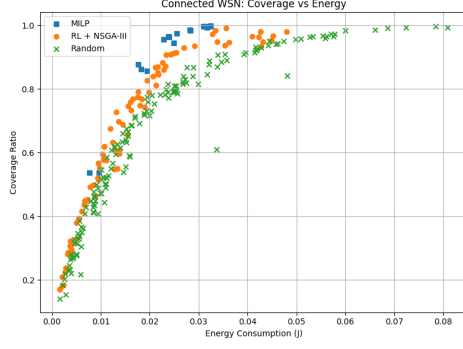
Example RL + NSGA-III Solution

Coverage = 0.9911, Energy = 0.035584 J, Hardware Cost = 340.68, Active Sensors = 29, Connected = True

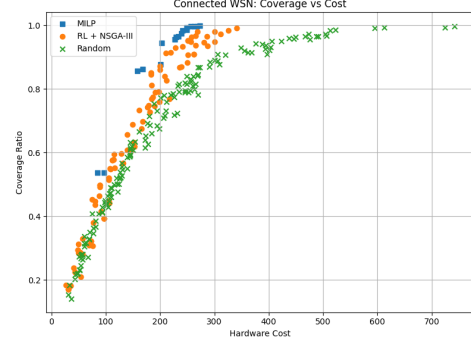
The three tables show a clear hierarchy in solution quality: The MILP weight-sweep provides the strongest overall Pareto approximation, reaching complete coverage while consistently requiring substantially fewer sensors, less energy, and lower hardware cost. In the reported experiment, all RL + NSGA-III and Random Search solutions are dominated by at least one MILP solution, confirming the superior resource efficiency of the exact optimization framework. Nevertheless, this advantage is obtained at an extremely high computational cost: MILP requires approximately hours, compared with to few minutes for RL + NSGA-III. For ample comparison details, refer to the comparison Figure A9, Table A50, Table A51, and Table A52.

Overall Comparison

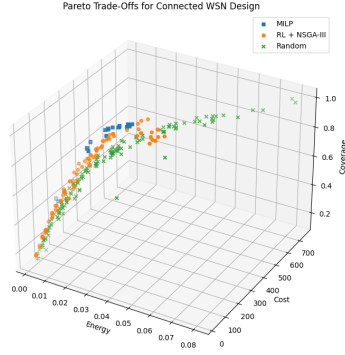
The three methods show a clear trade-off between solution quality and computational efficiency. Overall, MILP provides the best Pareto solution quality, achieving the highest coverage with the lowest energy, cost, and sensor count, but at a very high computational cost. RL + NSGA-III offers the best compromise, producing much better solutions than Random Search while being about 45.8× faster than MILP. Random



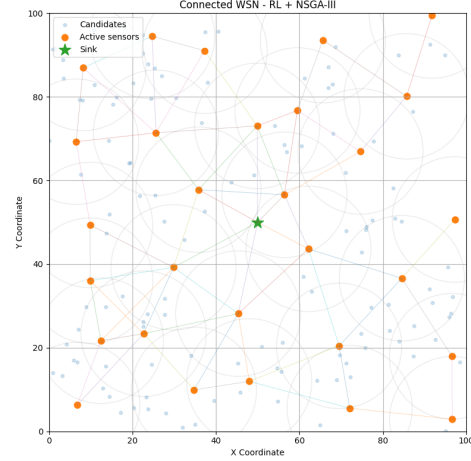
(a) Connected WSN: Coverage vs Cost



(b) Connected WSN: Coverage vs Cost



(c) Pareto Trade-Offs for connected WSN design



(d) Connected WSN coordinates - RL + NSGA-III

Fig. A9: Overall comparison of the four experimental results for seed=13, candidates=150.

Search is less competitive, requiring substantially more resources despite achieving high coverage.

Overall Interpretation

The results establish a clear ranking in terms of Pareto solution quality: MILP $\succ$ RL + NSGA-III $\succ$ Random Search. MILP provides the most resource-efficient solutions and, in this experiment, also achieves the highest coverage. However, this superiority comes at an extremely high computational cost, requiring more than 45 times the runtime of RL + NSGA-III. RL + NSGA-III therefore provides the most attractive quality-runtime compromise. Although its solutions are dominated by MILP in this run, it still produces substantially better configurations than Random Search while

Table A54: Overall Comparison with Seed=13-150

Criterion	MILP	RL + NSGA-III	Random Search
Maximum coverage	99.85%	99.11%	99.70%
Energy at maximum coverage	0.03247 J	0.03558 J	0.07840 J
Cost at maximum coverage	273.63	340.68	742.32
Sensors at maximum coverage	25	29	63
Execution time	8243.659 s	180.056 s	495.542 s
Connectivity	True	True	True

reducing MILP execution time by approximately 97.8%. Random Search can attain high coverage, but requires substantially more sensors, energy, and hardware cost. Overall, MILP is preferable when solution quality is the primary concern and computational time is acceptable, whereas RL + NSGA-III is the more practical approach when scalability and computational efficiency are important. Random Search remains useful primarily as a baseline rather than a competitive optimization method.

Overall Conclusion

Overall, MILP achieves the best solution quality and resource efficiency but is computationally expensive, whereas RL + NSGA-III provides a strong balance between solution quality and execution speed. Random Search is clearly less efficient and mainly serves as a baseline. Therefore, RL + NSGA-III represents the most practical scalable approach, while MILP remains preferable when computational cost is not a major constraint.

A.2.5 Sample output simulation results, seed=26 candidates = 150

RL + NSGA-III simulation profile is reported in Table A55:

Table A55: RL + NSGA-III simulation profile

Generation	Best Coverage	epsilon
0	0.9822	0.392
10	1.0000	0.320
20	0.9970	0.262
30	0.9985	0.214
40	1.0000	0.175
50	1.0000	0.143
60	0.9896	0.117
70	0.9882	0.095
80	0.9985	0.078
90	0.9970	0.064
100	0.9985	0.052
110	0.9985	0.042
120	0.9985	0.035
130	0.9970	0.028
140	0.9793	0.023

Table A56: MILP Pareto approximation

Coverage	Energy(J)	Cost	Sensors	Connected
1.0000	0.033305	269.58	25	True
1.0000	0.032361	278.69	26	True
1.0000	0.033511	267.96	25	True
0.9985	0.032849	264.07	25	True
0.9882	0.028437	256.96	23	True
0.9882	0.030768	240.34	23	True
0.9837	0.030110	233.74	22	True
0.9704	0.025150	238.95	21	True
0.9689	0.024541	242.40	21	True
0.9675	0.024513	238.99	21	True
0.9660	0.024761	233.24	21	True
0.9541	0.026941	203.81	20	True
0.9127	0.024358	178.22	18	True
0.8920	0.019387	197.76	17	True
0.8891	0.019444	192.83	17	True

Table A57: RL + NSGA-III Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9793	0.045426	267.62	24	True
0.9719	0.032981	268.24	23	True
0.9630	0.027618	246.54	21	True
0.9556	0.025817	233.76	20	True
0.9556	0.025817	233.76	20	True
0.9408	0.036932	224.91	19	True
0.9408	0.036932	224.91	19	True
0.9408	0.036932	224.91	19	True
0.9231	0.025171	239.96	20	True
0.9201	0.025745	225.95	19	True
0.9053	0.023395	227.65	19	True
0.9053	0.023395	227.65	19	True
0.9024	0.025774	203.10	18	True
0.8950	0.024130	201.33	18	True
0.8950	0.024130	201.33	18	True

Table A58: Random search Pareto front

Coverage	Energy(J)	Cost	Sensors	Connected
0.9985	0.079699	674.65	59	True
0.9985	0.078188	679.09	59	True
0.9970	0.075383	632.22	55	True
0.9941	0.051908	429.19	37	True
0.9778	0.050647	462.74	41	True
0.9778	0.051465	398.48	35	True
0.9719	0.050286	456.03	40	True
0.9689	0.046032	390.12	35	True
0.9586	0.051595	381.68	34	True
0.9571	0.050940	372.18	32	True
0.9571	0.046021	398.30	34	True
0.9497	0.037882	370.41	33	True
0.9482	0.052664	364.63	31	True
0.9438	0.051698	341.85	32	True
0.9438	0.045999	368.59	32	True

Table A59: Hypervolume Comparison

Method	Normalized Hypervolume	Relative to Random Search
MILP	0.5672	3.39×
RL + NSGA-III	0.4691	2.81×
Random Search	0.1672	1.00×

Referring to Table A59, MILP achieves approximately $1.21\times$ the hypervolume of RL+NSGA-III and $3.39\times$ that of Random Search. RL+NSGA-III achieves approximately $2.81\times$ the hypervolume of Random Search, corresponding to about a 180.6% larger dominated objective-space region.

Execution Time

MILP weight sweep : 55,606.984 s $\approx$ 15.45 hours

RL + NSGA-III: 191.259 s $\approx$ 3.19 minutes

Random Search: 376.011 s $\approx$ 6.27 minutes

The major disadvantage of MILP is its extremely large execution time: RL + NSGA-III is approximately $290.7\times$ faster than MILP, corresponding to a 99.66% reduction in execution time. It is also approximately $1.97\times$ faster than Random Search, or about 49.1% faster.

RL operator usage:

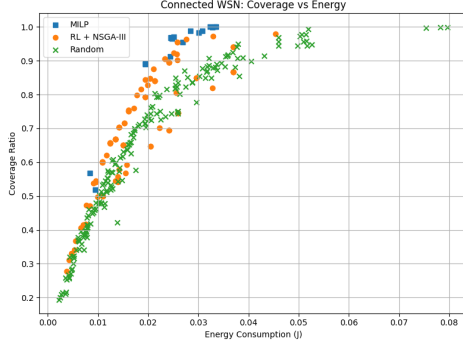
Bit Flip : 877, Add Sensor : 531

Remove Sensor : 2214, Swap Sensor : 719

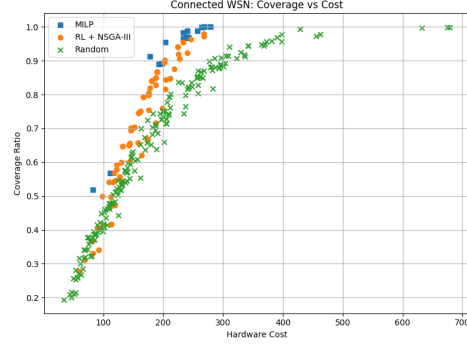
Coverage Guided : 10659

Example RL + NSGA-III Solution

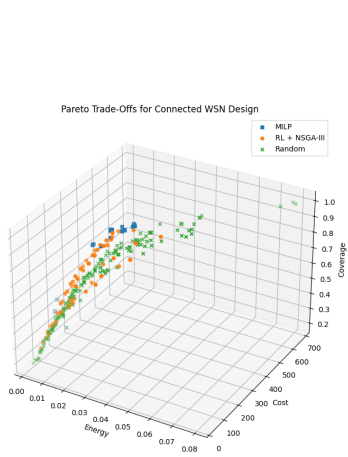
Coverage = 0.9793, Energy = 0.045426 J, Hardware Cost = 267.62, Active Sensors = 24, Connected = True



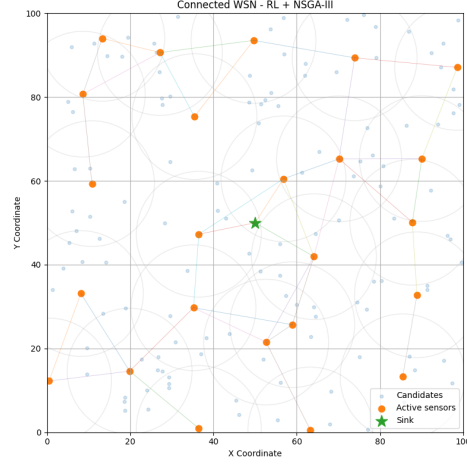
(a) Connected WSN: Coverage vs Cost



(b) Connected WSN: Coverage vs Cost



(c) Pareto Trade-Offs for connected WSN design



(d) Connected WSN coordinates - RL + NSGA-III

Fig. A10: Overall comparison of the four experimental results for seed=26, candidates=150.

The three tables show a clear hierarchy in solution quality: The MILP weight-sweep provides the strongest overall Pareto approximation, reaching complete coverage while consistently requiring substantially fewer sensors, less energy, and lower hardware cost. In the reported experiment, all RL + NSGA-III and Random Search solutions are dominated by at least one MILP solution, confirming the superior resource efficiency of the exact optimization framework. Nevertheless, this advantage is obtained at an extremely high computational cost: MILP requires approximately hours, compared with to few minutes for RL + NSGA-III. For ample comparison details, refer to the comparison Figure A10, Table A56, Table A57, and Table A58.

Overall Comparison

The three methods show a clear trade-off between solution quality/resource efficiency and computational cost. MILP generates the most resource-efficient Pareto solutions and is the only method reaching 100% coverage, but its execution time is extremely high. RL + NSGA-III achieves slightly lower maximum coverage but produces competitive solutions with much lower computational cost, running about $291\times$ faster than MILP and nearly $2\times$ faster than Random Search. Random Search can attain high coverage, but does so with substantially higher energy consumption, hardware cost, and sensor count. Overall, MILP is superior in Pareto solution quality, RL + NSGA-III provides the best balance between solution quality and computational efficiency, while Random Search is the least efficient approach overall.

Table A60: Overall Comparison with Seed=26-150

Criterion	MILP	RL + NSGA-III	Random Search
Maximum coverage	1.0000	0.9793	0.9985
Energy at maximum coverage	$\approx$ 0.033 J	0.0454 J	0.0782–0.0797 J
Cost at maximum coverage	$\approx$ 267.96–269.58	267.62	674.65–679.09
Sensors at maximum coverage	25–26	24	59
Connectivity	True	True	True
Execution time	55,606.984 s	191.259 s	376.011 s

Overall Interpretation

MILP provides the strongest Pareto approximation, achieving the best solution quality with fewer sensors, lower energy consumption, and reduced hardware cost, but at a very high computational cost of over 15 hours. RL + NSGA-III offers the best practical compromise, producing resource-efficient connected solutions within minutes and reducing execution time by about $291\times$ relative to MILP and 49% relative to Random Search. Random Search can reach high coverage but generally requires substantially more resources. Overall, MILP is preferable when solution quality is the main priority, while RL + NSGA-III is more suitable for computational efficiency and scalability; Random Search mainly serves as a weaker baseline.

Overall Conclusion

MILP provides the highest-quality and most resource-efficient Pareto solutions, but at a very high computational cost. RL + NSGA-III offers the best practical balance between solution quality and scalability, clearly outperforming Random Search while requiring far less computation than MILP. Overall, MILP is preferable when solution quality is the main priority, whereas RL + NSGA-III is more suitable for large-scale and time-sensitive WSN design.